

Financial Markets: 22. Public and Non-Profit Finance

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Chapter 1

Public and Non-Profit Finance

In this penultimate lecture Shiller deliberately widens the course. We are asked to stop thinking of finance as a narrow theory of securities and firms, and to see it instead as a general social technology for organizing incentives and allocating risks. Once we do that, nonprofit institutions, municipal borrowing, bankruptcy law, and social insurance no longer look like side topics. They become natural continuations of the same argument.

1.1 Finance Beyond the Business Firm

Shiller begins by restating, almost in compressed form, what the course has been about:

$$\text{Finance} \approx \text{incentives} + \text{risk management}. \quad (1.1)$$

That is not a theorem, but it is the governing formula of the lecture. Finance is about getting people to do good work together and about managing the risks that arise once human beings act in teams.

The lecture then pauses over a possible objection. Perhaps the greatest achievements are not team achievements at all. Perhaps they come from solitary genius. Shiller answers that objection by moving through a sequence of examples. Einstein may look like the lone thinker in a patent office, but even Einstein depends on education, journals, and a scientific community. Darwin may look self-directed, but the voyage of the *Beagle*, the collection of specimens, and the guidance of John Stevens Henslow all depend on institutions and patronage. Even Homer, imagined as the blind poet who simply remembers and recites, must still have had a world in which support, travel, transmission, and repetition were possible. The point is not to deny individual brilliance. It is to insist that brilliance becomes effective only inside some financing structure.

Only after that opening defense does the lecture announce its route. We are going to move through four domains:

- nonprofit organizations,
- government involvement in for-profit companies,
- government finance of projects, especially at the state and local level,
- government social insurance.

There is also a practical exhortation built into the opening. One should have a purpose in life other than making money per se, and finance should be thought of as a tool that helps us achieve that purpose. The lecture keeps returning to that thought.

1.2 Nonprofits as Mission-Driven Financial Forms

Shiller starts with the nonprofit because it is the clearest case of finance organized around an end other than private payout.

Definition 1.1. A nonprofit organization is a legal corporation whose charter states a purpose other than making money for owners. It has no shareholders in the ordinary sense, and any surplus is retained for the mission of the institution.

We can summarize that structure schematically as

$$\text{no shareholders, } \Pi_{\text{nonprofit}} \longrightarrow \text{organizational mission.} \quad (1.2)$$

The symbol $\Pi_{\text{nonprofit}}$ here means retained surplus, not distributable equity profit.

Yale is the lecture's immediate local example. It is a legal person, it has no owners, and it is governed by a board. The point is that the nonprofit form is not exotic. It is part of the daily institutional environment of the lecture itself.

Nor is it economically negligible. Shiller gives the lecture-time scale of the nonprofit sector as

$$N_{\text{U.S. nonprofits, 2010}} \approx 1.6 \times 10^6, \quad s_{\text{nonprofit in GDP}} \approx 4\%. \quad (1.3)$$

That is not the whole economy, but it is large enough that one must treat it as a serious part of financial life.

The lecture then turns from definition to example. Peter Tufano's Doorways to Dreams is presented as a nonprofit devoted to improving personal finance. One proposal is simple and behavioral: when a household receives a tax refund, the refund could be routed automatically into U.S. savings bonds rather than immediately taken in cash. Another proposal is even more revealing of the lecture's tone. Since many households are drawn to lotteries, why not design savings products with lottery-like random payouts, but without the expected loss that ordinary gambling creates? The form is nonprofit, but the logic is squarely financial. It uses incentive design to change saving behavior.

A second example is Dean Karlan's Innovations for Poverty Action, which the lecture presents at much larger scale:

$$\text{Innovations for Poverty Action income}_{2010} \approx \$25 \text{ million, } \text{staff} \approx 500. \quad (1.4)$$

Here the nonprofit form supports a global organization aimed at poverty alleviation. The point is not only that money can be raised. It is that the mission itself helps organize the raising of money.

Bill Drayton's Ashoka adds another element. The lecture treats it as a major institutional success and ties it to what it calls social entrepreneurship. The project is neither ordinary charity nor ordinary business rhetoric. It is organized around the idea that entrepreneurship can be directed toward social purpose.

Wendy Kopp then gives the youngest and sharpest example in the lecture. Teach for America begins as a senior thesis and becomes an institution:

$$\text{Teach for America initial fundraising} \approx \$2.5 \text{ million.} \quad (1.5)$$

That number matters because the idea itself was controversial. One could easily imagine resistance from established educational institutions or unions. Yet the nonprofit form allowed the project to seek financing directly, without first waiting for the government to adopt the idea as official policy.

1.2.1 Question & Answer

Question. Why would we choose a nonprofit rather than a for-profit form for the same broad social goal?

Answer. The lecture gives several answers at once. A nonprofit makes mission legible. Donors, banks, and counterparties can see more clearly that the organizer is not simply extracting private profit. That changes the atmosphere of fundraising. It also opens space for projects that are morally serious, politically unripe, or simply too strange to fit comfortably inside a government bureaucracy or an ordinary profit-maximizing firm.

The hospital stories then complicate the picture in exactly the right way. Yale New Haven Hospital is traced back to the General Hospital Society of Connecticut in 1826, and so to a nonprofit origin from the beginning. St. Raphael's is traced to physicians in New Haven who saw the need for a second hospital and worked with the Sisters of Charity. The natural question is the lecture's own question: why go to a religious nonprofit rather than simply create a for-profit hospital? Shiller's answer is cautious but clear. Affiliation with a religious mission gives moral identity to the organization and helps attract donations. Yet he immediately refuses a crude moral dichotomy. Studies comparing nonprofit and for-profit hospitals are, in his telling, generally inconclusive. Even in for-profit settings, people carry ideals and social purposes. The institutional boundary is real, but it is not morally absolute.

That ambiguity matters. It prepares the lecture for its next move. If the line between nonprofit and for-profit is already unstable on the nonprofit side, it will become even less stable when we look at government involvement in ostensibly private firms.

1.3 The Blurred Line Between Private Firms and Government

Shiller now turns explicitly to his second point: government involvement in for-profits. The lecture's claim is not merely that governments regulate firms from outside. It is that governments are already economically inside private enterprise through taxes, legal structure, and crisis management.

The first step is the corporate profits tax. Shiller interprets it as a standing public claim on private profit:

$$\text{government claim on corporate profit} = \tau_{\text{corp}}\Pi, \quad \text{private residual} = (1 - \tau_{\text{corp}})\Pi. \quad (1.6)$$

That is the clean mathematical core of his phrase “partial nationalization.” The government is not a common shareholder, but it does take a proportional claim on profits.

For the United States he gives the lecture-time arithmetic starkly:

$$\tau_{\text{corp,U.S.}} = 35\% + \tau_{\text{state/local}}, \quad \tau_{\text{state/local}} \leq 12\%, \quad \tau_{\text{corp,U.S.}} \leq 47\%. \quad (1.7)$$

So the total claim can be close to half of profit. He then broadens the comparison:

$$\tau_{\text{Canada}} \approx 16.5\% + 16\%, \quad \tau_{\text{Japan}} \approx 40.6\%, \quad \tau_{\text{Brazil}} \approx 34\%, \quad \tau_{\text{China}} \approx 25\%, \quad \tau_{\text{India}} \approx 33\%. \quad (1.8)$$

The lecture does not use these figures to update comparative public finance. It uses them to make a structural point: almost every country has such a tax, and no country can push it arbitrarily high without affecting the location of business activity.

1.3.1 Question & Answer

Question. In what sense is a private company partly public?

Answer. It is public in the sense that the state is already one of its claimants and guarantors. The firm is taxed, regulated, protected by legal rules, and, in some cases, rescued or restructured through public authority. So while it remains private in ownership and motive, it is not sealed off from public interest or public risk-bearing.

The TEPCO example makes the argument concrete. The lecture-time stock-price movement is summarized as

$$P_0^{\text{TEPCO}} \approx 25, \quad P_1^{\text{TEPCO}} \approx 5, \quad \frac{P_0^{\text{TEPCO}} - P_1^{\text{TEPCO}}}{P_0^{\text{TEPCO}}} = 0.80. \quad (1.9)$$

That is an 80% collapse. At the same time, estimated claims against the company are put at

$$\text{claims against TEPCO} \approx \$25 \text{ billion}. \quad (1.10)$$

The lecture's point is not merely that TEPCO had a disastrous event. It is that limited liability stops the liability of shareholders at the value of their equity, while the surrounding society still bears a great deal of the cost. A dispersed shareholder cannot inspect a nuclear plant, and the legal system does not ask ordinary investors to do so. That is precisely why the public claim through taxation, regulation, and emergency response is always in the background.

General Motors is the American counterpart. Here the lecture pivots through bankruptcy law. Chapter 7 means liquidation. Chapter 11 means reorganization: the company is unable to pay its current bills, but there remains something worth salvaging in continued operation. GM is treated as a Chapter 11 case because there is still a going concern to preserve.

The ownership restructuring that follows is one of the lecture's sharpest numerical moments:

$$\text{U.S. government} = 60.8\%, \quad \text{Canada} = 12\%, \quad \text{UAW} = 17.5\%. \quad (1.11)$$

The details of bondholders and warrants are noisier in the transcript, but the main result is clear enough: the new GM is, for a time, effectively government-owned. Old shareholders are wiped out; the restructured company survives inside a public-private settlement.

The lecture briefly adds another example of the same general phenomenon: personal bankruptcy. Here too the government is not merely observing private contracts from the outside. It is part of

the risk-management machinery that determines how far losses can propagate across a household lifetime.

At this point Shiller takes care not to let the chapter harden into a false taxonomy. He comes back to incentives inside nonprofits. The lecture-time statistic is

$$42\% \text{ of nonprofits pay bonuses to executives.} \quad (1.12)$$

So nonprofits use incentive contracts too. They must hire talent in a world where the for-profit sector also competes for talent. The government, by contrast, often faces public resistance to high compensation, and that can make recruitment harder. The underlying point is that incentive problems do not disappear when the mission becomes noble.

This is also where the lecture reintroduces the language of social entrepreneurs. Darwin's voyage, Wendy Kopp's educational idea, Ashoka, and similar projects all illustrate the same thought: some forms of progress are financed precisely because they stand between ordinary government action and ordinary profit. The world needs institutions that can fund weird, controversial, mission-driven projects.

1.4 State and Local Finance: Balanced Budgets, Capital Budgets, and Chapter 9

Shiller now moves to what he called the third thing: municipal or state and local finance. This is not a small administrative corner. State and local governments run schools, police and fire departments, parks, and much of the physical and civic infrastructure of everyday life. The lecture-time claim is that they spend roughly twice as much as the federal government. Whether or not one would restate the ratio today, the analytical point is plain: an enormous amount of social action runs through state and local finance.

The tone here is again practical. One can make things happen by founding a nonprofit, but one can also make things happen by approaching a local government with a project: a school, a bridge, a hospital, or some other durable public improvement. That is why the lecture turns next to the mechanics of local borrowing.

The apparent paradox is immediate. Every state in the United States, we are told, has some kind of balanced-budget rule. Yet cities and states borrow. Municipal debt obviously exists.

1.4.1 Question & Answer

Question. How can a government have a balanced-budget rule and still borrow heavily?

Answer. Because the rule is aimed primarily at the operating budget. Long-lived projects are pushed onto a distinct capital budget, and that budget can be financed with debt. So the contradiction disappears once we separate current operations from durable investment.

A compact formalization is

$$T_t \approx E_t^{\text{op}}, \quad (1.13)$$

$$E_t^{\text{cap}} \text{ may be financed by } \Delta B_t > 0. \quad (1.14)$$

Here T_t is current tax revenue, E_t^{op} operating expenditure, E_t^{cap} capital expenditure, and B_t debt. The balanced-budget discipline applies to current operations, not to the entire intertemporal plan of the jurisdiction.

The new-town example is the lecture's clearest worked exercise in public finance. Suppose a town is tiny now but expects to grow substantially:

$$\text{current population} \approx 30, \quad \text{planned population} \approx 20,000, \quad \text{sewer cost} \approx \$100 \text{ million.} \quad (1.15)$$

A sewage engineer tells us that the system should be built in one coherent plan, not extended piecemeal house by house. The town therefore faces a lumpy capital project whose benefits lie mostly in the future.

The logic unfolds in a few steps:

1. The current tax base is far too small to fund the project out of current operating revenue.
2. The sewer system must nevertheless be built at scale in advance if the future city is to function properly.
3. So the expense belongs on the capital budget rather than the operating budget.
4. Debt is then serviced by the larger future tax base that the infrastructure was built to support.

That is why the lecture calls the arrangement fair and just. It aligns the burden of payment more closely with the distribution of benefit across time. But the darker side follows immediately. What if the forecast is wrong? What if the people never come? Then the city has borrowed against a future that never materializes, and the debt becomes a source of distress rather than a bridge to growth.

This is where Chapter 9 enters. Unlike a corporation, a city has no shareholders. It cannot simply liquidate equity. It does have taxing power, but taxing power is not absolute, because excessive taxation drives residents and businesses away. Municipal bankruptcy is therefore a subtler problem than corporate bankruptcy. Solvency depends on preserving the economic and political life of the city itself.

Shiller notes that municipal bankruptcies had historically been rare, though not impossible. New York City in the 1970s is the central example of near-failure without formal bankruptcy, largely because state and federal bailouts intervened. Cities also try to self-insure through rainy day funds, accumulating reserves in better times to smooth bad periods. But the lecture is speaking in the shadow of the financial crisis, when many such reserves had already been exhausted. That is the immediate backdrop for the next move: rising concern about municipal distress translated into rising municipal yields.

1.5 Municipal Bonds, Tax Exemption, and Yale's Borrowing Logic

At this point the lecture makes its sharpest pricing turn. Municipal distress is no longer just a legal possibility; it begins to enter market valuation.

The blackboard here is not algebra; it is a sequence of headings. Read cautiously, and with the transcript supplying the completion of the last phrase, the board gives us

$$\text{Chapter 9,} \quad \text{Rainy Day Fund,} \quad \text{Tax Free.} \quad (1.16)$$

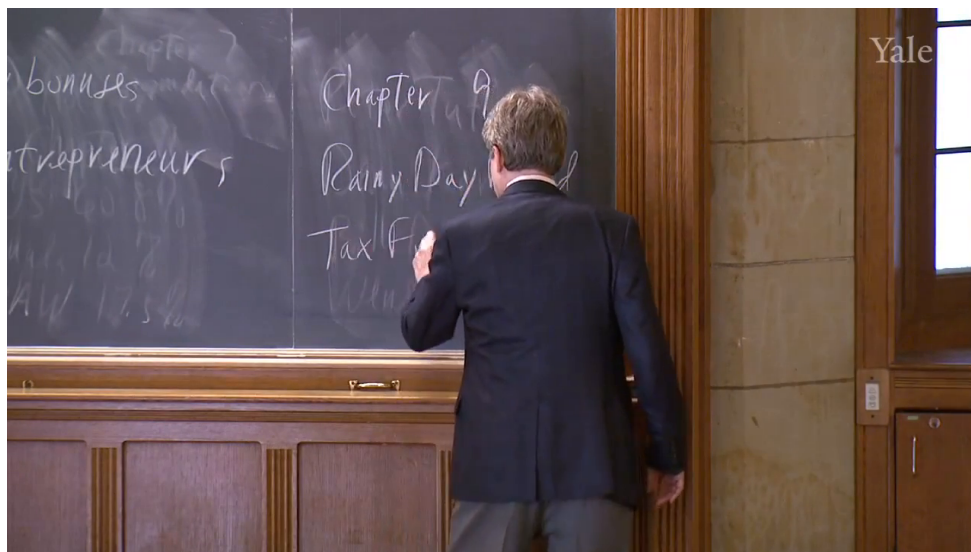


Figure 1.1: Chapter 9, rainy day funds, and tax-free municipal bonds

That is exactly the order of the lecture's reasoning at this point: bankruptcy risk, reserve accumulation, and then tax treatment. The faint words on the left belong to the earlier nonprofit discussion and do not belong to the municipal-bond argument.

The lecture's clean financial statement is then:

$$\text{Interest on municipal bonds is exempt from federal income tax.} \quad (1.17)$$

In the lecture's own telling, that tax treatment reflects the constitutional separation between federal and state authority. Whether we emphasize the constitutional detail or simply the tax rule, the pricing implication is the same: municipal bonds carry a tax subsidy.

We can write the tax argument in the standard yield language. Let y_{muni} be the yield on a municipal bond and let τ_{inv} be the investor's marginal tax rate. If a taxable bond is to leave the investor indifferent, its after-tax yield must match the municipal yield:

$$(1 - \tau_{\text{inv}}) y_{\text{taxable-equivalent}} = y_{\text{muni}}, \quad (1.18)$$

$$y_{\text{taxable-equivalent}} = \frac{y_{\text{muni}}}{1 - \tau_{\text{inv}}}. \quad (1.19)$$

This is the precise version of Shiller's point that high-income investors especially value municipal bonds. The higher the tax rate, the more attractive the exemption.

The lecture also notes that municipal yields had risen. There is no contradiction. Tax exemption pulls yields down; default or restructuring risk pushes them up. Municipal pricing is therefore the combination of a tax wedge and a credit premium.

1.5.1 Question & Answer

Question. Why would Yale borrow despite having a huge endowment?

Answer. Because Yale can borrow on tax-favored terms. Once debt is tax-exempt, the borrowing rate is reduced, and debt can become attractive even for a wealthy institution. That, in turn, creates the temptation to compare a low tax-exempt borrowing cost with the return on higher-yield assets.

The lecture's arithmetic is:

$$\text{endowment} = \$16 \text{ billion}, \quad \text{debt} = \$2.5 \text{ billion}, \quad 16 - 2.5 = 13.5. \quad (1.20)$$

At first glance one might say: why borrow at all if one has \$16 billion? But the lecture's immediate answer is financial. If Yale can issue tax-exempt debt, then it can borrow at a relatively low rate. That invites the comparison

$$\text{apparent spread} = r_{\text{taxable asset}} - r_{\text{tax-exempt debt}}. \quad (1.21)$$

Since Yale is tax-exempt on the asset side as well, one sees why the spread looks tempting. Shiller explicitly describes this as an arbitrage game that a nonprofit might wish to play.

But he also immediately limits the thought. This is not an unrestricted free lunch. A nonprofit cannot simply borrow without purpose and lever itself into any taxable asset it likes. The proceeds of tax-favored borrowing must be tied to proper institutional uses. The lecture's point is not that Yale is engaging in boundless arbitrage. It is that the tax treatment makes indebtedness intelligible rather than puzzling.

Shiller closes this municipal-finance part with one more institutional comparison. State governments typically distinguish operating and capital budgets. The federal government, in his lecture-time description, has only one budget and no separate capital budget. In that narrow sense the state and local framework can look more explicit than the federal one about the difference between current operations and long-lived investment.

1.6 Social Insurance and the Technology of Administration

The lecture then moves to its last topic and widens again. Finance, we are reminded, is about risk management and incentivization. Social insurance is one of the largest institutional forms in which those functions are performed.

Definition 1.2. Social insurance is government-provided insurance against major adverse states of life or income, especially in areas where private insurance is incomplete, unavailable on acceptable terms, or structurally difficult to provide.

Shiller begins with progressive taxation. Properly stated, progressive taxes impose higher marginal rates on higher incomes. But the lecture's real emphasis is the Earned Income Tax Credit. For sufficiently low earned income, the net tax schedule can become negative:

$$T(y) < 0 \quad \text{for part of the low-income range under the EITC.} \quad (1.22)$$

That means a fall in pretax income is only partly transmitted to after-tax income. The tax system itself becomes a risk-sharing device.

The same thought extends to public services. Education, in particular, is treated in the lecture as a kind of social insurance. It guarantees access to a foundational good even when private earning

power is weak. The lecture's claim is that such services protect people against failure in the economic system itself.

Shiller then turns to the classic U.S. social-insurance cluster, which we may label more carefully as OASDI: old-age, survivors, and disability insurance. The components differ, but the financial logic is shared:

- old-age insurance provides pension income in retirement,
- survivors' insurance provides support to dependents after the death of a parent or wage earner,
- disability insurance provides income when permanent impairment makes work impossible.

We can compress that intertemporal structure as

$$\text{current contributions} \longrightarrow \text{future pension} / \text{survivor} / \text{disability benefits.} \quad (1.23)$$

Health insurance and workers' compensation then complete the list. Workers' compensation is described as an older form of insurance against accidents at work. Health insurance is discussed more hesitantly through the U.S. case, precisely because the United States is not presented as the world leader in that domain. The lecture-time number is

$$\text{uninsured}_{\text{U.S.}} \approx 40 \text{ million.} \quad (1.24)$$

Private markets do overlap with these programs. One can buy pensions, life insurance, and disability insurance privately. But the lecture insists that private provision is not enough. Disability insurance, in particular, suffers from adverse selection: people who suspect they are more likely to become disabled have a stronger incentive to buy the policy. That is one reason the state enters.

1.6.1 Question & Answer

Question. Why did social insurance require modern information technology and administration?

Answer. Because social insurance is not merely compassion backed by taxation. It is a long-lived accounting system. Someone must identify people, record contributions, preserve records, verify claims, transmit documents, and standardize procedures over long stretches of time and over large territories.

The lecture makes this historical claim through Bismarckian Germany in the 1880s. That is where national social insurance first takes hold at scale. Shiller gives two dimensions of the administrative burden:

$$\text{workers in Bismarck's system} \approx 11 \text{ million,} \quad H_{\text{record-keeping}} \approx 30 \text{ to } 40 \text{ years.} \quad (1.25)$$

Those numbers make clear why observers in London thought the scheme would collapse. A pension system requires the state to keep a tab on contributions over decades and then to pay benefits later on the basis of those records. Without a reliable bureaucracy, the promise is empty.

The lecture reinforces the point by contrast. Shiller mentions an early English wage-supplement experiment, commonly associated with Speenhamland, in which the town promised to make up the difference between actual earnings and a deemed living income. The scheme failed because

administrators could not verify true income or effort. People could not be monitored accurately enough.

What changed in the nineteenth century? The lecture answers with a list that at first sounds prosaic and then reveals itself as decisive: cheap paper, carbon paper, typewriters, standardized forms, filing cabinets, management schools, bureaucracy, and the postal system. These are not ornaments around finance. They are what make large-scale contingent claims administratively possible. The image of workers going to local post offices, making payments, receiving stamps, and pasting them onto cards is exactly the sort of detail the lecture wants us to notice. Information technology does not merely improve finance. It creates new domains of finance.

That is why the lecture ends with a larger claim: technology drives finance, and finance changes when technology changes. Social insurance is one of the clearest historical examples.

1.7 Summary

The lecture has moved through nonprofits, private firms, municipalities, and national social insurance, but the underlying structure has remained the same. Finance is the organization of incentives and the sharing of risk. Nonprofits make mission legible and finance projects that may be too controversial or too purpose-driven for ordinary public administration. Corporate taxes, bankruptcy law, and bailouts show that private firms are never fully outside the public sphere. State and local borrowing converts future tax bases into present infrastructure. Municipal bonds combine tax privilege with credit risk. Social insurance extends the same logic to the scale of a whole society.

What remains at the end is a broadened view of the subject. Finance is not an end in itself. It is a set of institutional devices through which people pursue purposes, attract capital, motivate work, and survive bad states of the world. That is why the distinction between for-profit, nonprofit, and government activity is real but never complete. All three are part of the same civilizational effort to align incentives, ideals, and administrative capacity.